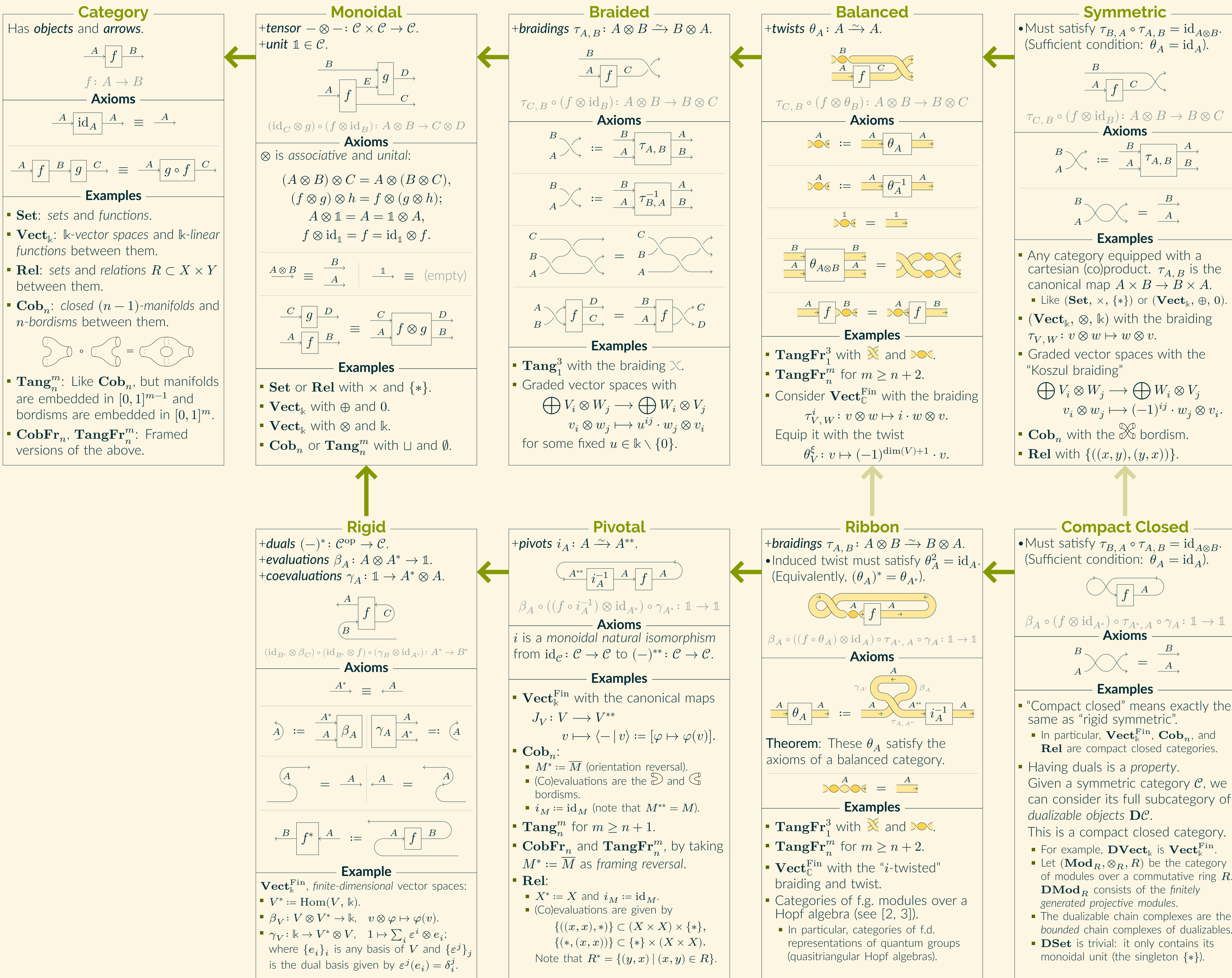


A brief introduction to graphical calculus

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An example: dimension and trace

- Let $\mathcal{C}$ be a ribbon category.
- Note that $\text{End}_{\mathcal{C}}(1)$ is a *monoid* under composition. We can regard its elements as “numbers”. (Usually we work over **Mod_R**-enriched categories, which makes $\text{End}_{\mathcal{C}}(1)$ into an R -algebra).
- We define the **categorical trace** of an endomorphism $f: A \rightarrow A$ as the composition

$$\text{tr}(f) := \text{tr}(f) \in \text{End}_{\mathcal{C}}(1).$$

- We define the **categorical dimension** of an object A as $\text{tr}(\text{id}_A)$:

$$\dim(A) := \text{tr}(\text{id}_A) \in \text{End}_{\mathcal{C}}(1).$$

- The trace is symmetric; $\text{tr}(g \circ f) = \text{tr}(f \circ g)$. We can prove it via the graphical calculus:

$$\begin{aligned} \text{tr}(g \circ f) &:= \text{tr}(g \circ f) = \text{tr}(f \circ g) = \text{tr}(f \circ g) = \text{tr}(g \circ f) \\ &= \text{tr}(g \circ f) = \text{tr}(f \circ g) = \text{tr}(g \circ f) = \text{tr}(f \circ g) \end{aligned}$$

- Working over **Vect_k^{Fin}**, we recover the usual trace:

$$\begin{aligned} \mathbb{k} &\xrightarrow{\gamma_V} V^* \otimes V \xrightarrow{\tau_{V^*, V}} V \otimes V^* \xrightarrow{(f \otimes \text{id}_{V^*}) \circ \text{id}_{V^*}} V \otimes V^* \xrightarrow{\beta_V} \mathbb{k} \\ 1 &\mapsto \sum_i \varepsilon^i \otimes e_i \mapsto \sum_i e_i \otimes \varepsilon^i \mapsto \sum_i f(e_i) \otimes \varepsilon^i \mapsto \sum_i \varepsilon^i(f(e_i)) = \text{tr}(f). \end{aligned}$$

- In the category of (dualizable) R -modules, the dimension $\dim(M)$ recovers the *rank* $\text{rank}_R(M)$.
- In the category of (dualizable) chain complexes of R -modules, when equipped with the “Koszul braiding”, the dimension $\dim(\mathcal{C}_\bullet)$ recovers the *Euler characteristic* $\chi(\mathcal{C}_\bullet) = \sum_i (-1)^i \text{rank}_R(\mathcal{C}_i)$.

An application: Reshetikhin–Turaev invariants of 3-manifolds

Very broad sketch of the construction. See [2, 3] for details.

- Let $\mathcal{C}$ be a ribbon category.
- By the graphical calculus, we have an assignment
 $\left\{ \mathcal{C}\text{-labelled framed tangles in } \mathbf{TangFr}_1^3(\mathcal{C}) \right\} \rightarrow \left\{ \text{Morphisms in } \mathcal{C} \right\}$
given by evaluating a diagram of the tangle in our category. (This is an *isotopy invariant*).
- In particular, we have an assignment
 $\left\{ \mathcal{C}\text{-labelled framed links} \right\} \rightarrow \text{End}_{\mathcal{C}}(1).$

(Under some specific choices of $\mathcal{C}$ and labels, this recovers the *Jones polynomial* of each link).

- Under some conditions on $\mathcal{C}$, we can extend this assignment to *unlabelled* framed links:

$$\left\{ \text{Framed links} \right\} \xrightarrow{\text{RT}_{\mathcal{C}}} \text{End}_{\mathcal{C}}(1).$$

Basically, we label each component of our link by the simple objects of $\mathcal{C}$ in all the possible ways. This yields a finite family of invariants in $\text{End}_{\mathcal{C}}(1)$. We define the **Reshetikhin–Turaev invariant** of the framed link L , $\text{RT}_{\mathcal{C}}(L)$, as a certain weighted sum of this family of invariants.

(This step requires $\mathcal{C}$ to be a *modular tensor R -category*, which consists of a semisimplicity condition and a finiteness condition. Additionally, $\mathcal{C}$ is enriched over R -modules, so $\text{End}_{\mathcal{C}}(1)$ is an R -algebra).

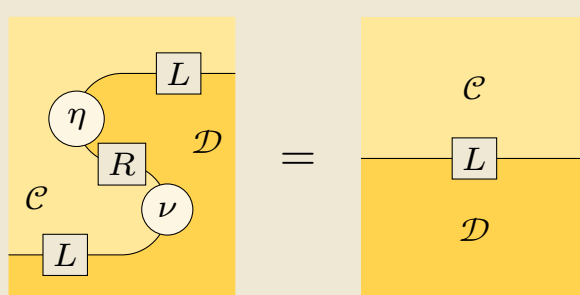
- By the Lickorish–Wallace theorem, any closed oriented 3-manifold M can be obtained by Dehn surgery on the 3-sphere along a framed link L . We define the **Reshetikhin–Turaev invariant** of the 3-manifold M as $\text{RT}_{\mathcal{C}}(M) := \text{RT}_{\mathcal{C}}(L)$. This does not depend on the choice of link used.

$$\left\{ \text{Closed oriented 3-manifolds} \right\} \xrightarrow{\text{RT}_{\mathcal{C}}} \text{End}_{\mathcal{C}}(1).$$

- Additional work makes this into a 3D TQFT $Z_{\mathcal{C}}: \mathbf{Cob}_3^{\mathcal{C}} \rightarrow \mathcal{C}$.
By seeing a closed 3-manifold M as a bordism $\emptyset \rightarrow \emptyset$, we recover $\text{RT}_{\mathcal{C}}(M) = Z_{\mathcal{C}}(M) \in \text{End}_{\mathcal{C}}(1)$.

Bicategories

- Similar diagrams, but we color the regions to denote 0-cells (objects).
- Consider the bicategory **Cat** of (small) categories, functors, and natural transformations. This diagram depicts one half of the definition of adjointness of functors $\mathcal{C} \xrightleftharpoons[R]{L} \mathcal{D}$.
 - $\mathcal{C}$ and $\mathcal{D}$ are categories (objects).
 - $L: \mathcal{C} \rightarrow \mathcal{D}$ and $R: \mathcal{D} \rightarrow \mathcal{C}$ are functors (1-morphisms).
 - The unit $\eta: \text{id}_{\mathcal{C}} \Rightarrow R \circ L$ and counit $\nu: L \circ R \Rightarrow \text{id}_{\mathcal{D}}$ are natural transformations (2-morphisms).



- In *monoidal* bicategories, the diagrams become 3D.
- This diagram depicts one half of the definition of 2-dualizable object.

$$\begin{aligned} \beta_A &\Downarrow \text{id}_B \otimes \eta \\ \beta_A \circ \beta_A &\Downarrow \nu \otimes \text{id}_B \\ \beta_A &\Downarrow \beta_A \end{aligned} \quad \text{Diagram} = \quad \text{Diagram}$$